CHUN-HAO TO

US Legal Permanent Resident (Green Card holder)

CONTACT

William Eckhardt Research Center
5640 S Ellis Ave, Room 413
Chicago, IL, 60637

E-Mail: chto@uchicago.edu
Phone: +1-702-235-8760
Webpage: chunhaoto.com

RESEARCH INTERESTS

Observational and Computational Cosmology
Weak gravitational lensing, Large-scale structure, Cosmological simulation, Cluster cosmology, Galaxy–halo connection, Statistical and AI methods for robust inference

EMPLOYMENT

Schmidt AI in Science Fellow, University of Chicago *2024 – present*
CCAPP Postdoctoral Fellow, the Ohio State University *2021–2024*

EDUCATION

Ph.D. in Physics *Sep 2016– Sep 2021*
Department of Physics, Stanford University, CA, USA
Advisor: Risa H. Wechsler. Thesis: "Multi-probe cluster cosmology analyses with photometric surveys."

B.S. in Physics *Sep 2011– June 2015*
Department of Physics, National Taiwan University, Taipei, Taiwan

GRANTS AND ALLOCATIONS

- Roman Cycle 1 Program: A New Look at Dark Matter in the Roman Era: Joint Inference of High-Resolution Mass Map and Cosmology [co-PI] *2026*
- Midway Research II allocation, 2M hours [PI] *2024,2025,2026*

AWARDS

- Schmidt AI in Science Fellowship *2024*
- CCAPP Fellowship *2021*
- Dark Energy Survey builder status (for 2 FTE years of infrastructure work) *2020*
- Dark Energy Survey Early Career Scientist *2019*
  Awarded US$3500 for participating in 4 DES Collaboration Meetings
- Dean's Award of College of Science, National Taiwan University *2015*
- Presidential Awards, National Taiwan University *2011-2015*

SCIENCE LEADERSHIP

- Inaugural co-chair of the catalog and statistics working group in the Roman High Latitude Imaging Survey Project Infrastructure Team (2026–)
  - Serve on the HLIS-PIT steering committee, helping define the team's science goals and analysis priorities.
  - One of the three working groups in the Roman team that builds the infrastructure for weak lensing analyses.
  - Coordinate tasks including photometric redshift estimation and characterization, galaxy sample selection, cluster detection, catalog simulations, data vector generation, and parametrization of observational systematics.
- Co-chair of the survey working group in the Roman High Latitude Imaging Survey Project Infrastructure Team (2025–2026)
  - One of the three working groups in the Roman team that builds the infrastructure for weak lensing analyses.
- Co-convenor of the cluster working group in the Dark Energy Survey (2024–)
  - Designed, led, and delivered the year three cluster cosmology constraints.
  - Co-initiated the simulation-based cluster cosmology key project.
- Key project coordinator of the cluster working group in the Dark Energy Survey (2021–2024)
  - Led the development of simulations used for the full Dark Energy Survey key project analyses.
  - Led the development of the likelihood sampling tool for the joint analysis of clusters, weak lensing, and large-scale structure.

SERVICE

- Professional Service
  - Roman Reprocessing & Release Coordinating Committee.
  - Grant reviewer for: NASA Astrophysics Data Analysis Program Grant (ADAP).
  - Journal referee for: MNRAS, MNRAS-Letter, APJ, APJS, and A&A.
- Departmental Service
  - Open SkAI conference LOC, 2025.
  - Organizer of Schmidt AI summer school at UChicago, 2025.
  - Organizer of SkAI journal club at UChicago, since 2025.
  - Organizer of Schmidt AI in science seminar series at UChicago, since 2024.
  - Organizer of Cosmolunch at OSU, 2021-2024.
- Collaboration Service
  - Roman HLIS-PIT WG chairs selection committee, 2026.

- SOC of Roman HLIS-PIT collaboration meeting, 2026.
- Internal referee of the Dark Energy Survey, since 2020.
- Internal referee of the Dark Energy Spectroscopic Instrument, 2025.
- Observing shift for the Dark Energy Spectroscopic Instrument, 2021

TEACHING AND OUTREACH

- Astronomy Conversations Volunteer at Adler Planetarium *Fall 2024-*
- Judge of South Side Science Astronomy Art Contest *2025*
- Public talk at Bexley library:
  "Simulating the Universe with Supercomputers" *Summer 2023*
  "Learning the dark universe with galaxies and galaxy clusters" *Summer 2022*
- KIPAC Blog Post Exploring the Cosmos while Preserving Spaceship Earth *Fall 2022*
- Guest lecturer at Astronomy 2895 Topics in Astrophysics, OSU *Fall 2021*
- Guest lecturer at Physics 367: Structure formation and galaxy evolution, Stanford *Spring 2021*
- Teaching assistant for The Origin and Development of the Cosmos, Stanford *Winter 2020*
- Teaching assistant for Astronomy Laboratory and Observational Astronomy, Stanford *Fall 2018*
- Teaching assistant for Electricity and Magnetism Lab, Stanford *Spring 2017*
- Teacher, Stanford ESP Splash! Program, Stanford *Spring 2017*

MENTORING

I have played a significant role in supervising research projects of the following students.

- Yufei Zhen, master student at the Data Science Institute of UChicago, Self-organizing map-based photometric redshift estimations of Roman Core Community Survey, 2025–
- Nihar Dalal, graduate student at the OSU, co-supervising with Chris Hirata, Constraining baryonic physics with ACT clusters and DES weak lensing, 2023–
- I-Hsuan Li, visiting undergrad student at NCKU, cluster finding with Roman and Rubin, 2025
- Gwen Sheley, undergrad student at UChicago, Quantify systematics in tSZ x shear, UChicago Quad Undergraduate Research Scholar, 2024–2025
- Kathlynn Simotas, undergrad student at Stanford (past), UCSB graduate student, Quantifying redMaPPer cluster systematics using galaxies with spectroscopic redshifts, 2019–2021

PRESS

Scientists release new survey of the biggest objects in the universe [UChicago 2025]

Dark Energy Survey physicists open new window into dark energy [SLAC 2021]

A New View of the Universe's Dark Side [APS 2021]

PROGRAMMING

Extensive experience in Python, PyTorch, C/C++, JAX, and IDL.

GitHub Page: `https://github.com/chto`.

HLIS-PIT pipelines: Photometric redshift estimation and Photometry measurement.

PRESENTATIONS

* Invited Presentations

2026
- Lunch Talk, ASIAA *Sep*
- New Cross-Correlation Perspectives on Cosmic Baryons at Aspen Physics Center *Aug*
- *NASA Cosmic Structure Science Interest Group Seminar *June*
- Rubin/Roman Joint Processing at KIPAC *May*
- Roman Research Discussion at Goddard Space Flight Center *May*
- *Princeton/IAS Cosmology Seminar *April*
- Roman Quarterly (Virtual): HLIS-PIT's plan on year one analysis *April*
- *MockNYC 2026: Improving Galaxy Modeling for Cosmological Inference *Jan*

2025
- Cosmo 25: 28th International Conference on Particle Physics & Cosmology, Pittsburgh, PA *Oct*
- Observers' group seminar, CIERA, Northwestern University *Sep*
- *Tracing Cosmic Evolution with Galaxy Clusters V, Sexten Center for Astrophysics Riccardo Giacconi *July*
- *mm Universe 2025, KICP *June*
- *Colloquium, Universität Innsbruck *April*
- *Colloquium, Ludwig Maximilian University of Munich *April*
- *APS Global Physics Summit 2025, Exploring the Expanding Universe with the Dark Energy Survey *March*

2024
- Cosmology and galaxy astrophysics with simulations and machine learning, CCA *Dec*
- Roman HLIS PIT F2F meeting, IPAC *Oct*
- Seminar, LMU *May*
- *Colloquium, ASIAA *March*

2023
- *Cosmology Astrophysics Seminar at University of Michigan *Nov*
- Plenary talk at Dark Energy Survey Collaboration Meeting *Oct*

	• Groups and Clusters of Galaxies at the Crossroad between Astrophysics and Cosmology at Aspen	*Aug*
	• *CMB-S4 Workshop	*Aug*
2022	• ICML International Conference on Machine Learning	*July*
	• Plenary talk at Dark Energy Survey Collaboration Meeting	*May*
	• *Colloquium, LMU	*May*
	• Colloquium, ASIAA	*May*
	• Advances in Cosmology through Numerical Simulations	*May*
	• Galaxy Clusters 2022: Challenging Our Cosmological Perspectives	*April*
2021	• Cluster Mass 2020	*Sep*
	• Colloquium, ASIAA	*Jan*
2020	• Survey science group meeting, University of Chicago	*Nov*
	• *Gravitational lensing group seminar, LMU	*Nov*
	• *Seminar at Stony Brook University	*Nov*
	• *Dark Sector Meeting, JPL	*Oct*
	• Cosmology Lunch, Princeton University	*Oct*
	• Cosmic surveys meeting, Fermilab	*Oct*
	• *CCAPP Seminar, OSU	*Oct*
	• *Cosmology Seminar, University of California, Berkeley	*Sep*
	• *11th CMB-S4 Workshop: Cosmology and Astrophysics in the Next Decade	*Aug*
	• Cosmology from Home	*Aug*
	• Dark Energy Survey Virtual Collaboration Meeting	*May*
2019	• Dark Energy Survey Collaboration Meeting	*Nov*
	• Cosmic Controversies	*Sep*
	• Great Lake Cosmology Workshop	*Aug*
	• Dark Energy Survey Collaboration Meeting	*May*
	• Panchromatic Panoramic Studies of Galaxy Clusters	*March*
2018	• Dark Energy Survey Y3KP workshop	*Oct*
	• Tucson DECaLS Workshop	*Aug*
	• Santa Cruz Galaxy Workshop	*Aug*

PUBLICATIONS

190 refereed publications with a total of 13146 citations (12735 citations in refereed Journals), h-index=60 according to NASA/ADS Metrics Summary.
Note that there are three separate numbered sections below: **Leading contributions** (main contributor of the project or papers from students I directly advise and have significant contributions) of 15 publications in lensing cosmology, simulations, cross-correlations, cluster cosmology, and other large-scale structure cosmology topics, **Supporting contributions** of 26 papers with a brief description of the contributions, and **Selected publications that are credited as a DES builder**. Full publication list available at ADS.

- **Leading contributions:**
  * **Student-led papers.**

15. Anbajagane, D., **To, C.-H.**, Gaido, M., et al. 2026, "A DECADE of Giants: A catalog of 42,000 optically identified galaxy clusters from 13,000 deg^2 of the sky", *arXiv e-prints*, arXiv:2610.24104

14. *Zhen, Y., **To, C.-H.**, Yin, B., et al. 2026, "Extending SOMPZ beyond Spectroscopic Coverage for Roman Weak Lensing", *arXiv e-prints*, arXiv:2610.24066

13. **To, C.-H.**, Shin, T.-H., Chang, C., et al. 2026, "Robust Photometry for Roman High-Latitude Imaging Survey Cosmology Using Roman and Rubin Imaging", *arXiv e-prints*, arXiv:2607.09849

12. **To, C.-H.**, Chang, C., Anbajagane, D., et al. 2026, "First detection of weak lensing around spectroscopically confirmed low-mass galaxies", *Physical Review D*, 114, 023040

11. *Dalal, N., **To, C.-H.**, Hirata, C., Shin, T.-H., Hilton, M., Pandey, S., & Bond, J. R. 2026, "Deciphering baryonic feedback from ACT tSZ galaxy clusters", *Journal of Cosmology and Astroparticle Physics*, 2026, 036

10. Abbott, T. M. C., Aguena, M., Alarcon, A., et al. 2025, "Dark energy survey year 3 results: Cosmological constraints from cluster abundances, weak lensing, and galaxy clustering", *Physical Review D*, 112, 083535

    Alphabetically ordered. Contributions include Scientific management and coordination, Project development (including paper writing and figures), Data analysis and methods validation, and Data vector generation. See contributions in the Author Contributions section.

9. **To, C.-H.**, Krause, E., Chang, C., et al. 2025, "Dark energy survey: Modeling strategy for multiprobe cluster cosmology and validation for the full six-year dataset", *Physical Review D*, 112, 063537

8. Bocquet, S., Grandis, S., Krause, E., **To, C.-H.**, et al. 2025, "Multiprobe cosmology from the abundance of SPT clusters and DES galaxy clustering and weak lensing", *Physical Review D*, 111, 063533
   The second to the fourth authors are alphabetically ordered.

7. **To, C.-H.**, Pandey, S., Krause, E., Dalal, N., Anbajagane, D., & Weinberg, D. H. 2024, "Deciphering baryonic feedback with galaxy clusters", *Journal of Cosmology and Astroparticle Physics*, 2024, 037

6. **To, C.-H.**, DeRose, J., Wechsler, R. H., et al. 2024, "Buzzard to Cardinal: Improved Mock Catalogs for Large Galaxy Surveys", *The Astrophysical Journal*, 961, 59

5. **To, C.-H.**, Rozo, E., Krause, E., Wu, H.-Y., Wechsler, R. H., & Salcedo, A. N. 2023, "LINNA: Likelihood Inference Neural Network Accelerator", *Journal of Cosmology and Astroparticle Physics*, 2023, 016

4. **To, C.-H.**, Krause, E., Rozo, E., et al. 2021, "Dark Energy Survey Year 1 Results:

Cosmological Constraints from Cluster Abundances, Weak Lensing, and Galaxy Correlations", *Physical Review Letters*, 126, 141301

3. **To, C.-H.**, Krause, E., Rozo, E., et al. 2021, "Combination of cluster number counts and two-point correlations: validation on mock Dark Energy Survey", *Monthly Notices of the Royal Astronomical Society*, 502, 4093

2. **To, C.-H.**, Reddick, R. M., Rozo, E., Rykoff, E., & Wechsler, R. H. 2020, "RedMaPPer: Evolution and Mass Dependence of the Conditional Luminosity Functions of Red Galaxies in Galaxy Clusters", *The Astrophysical Journal*, 897, 15

1. **To, C.-H.**, Wang, W.-H., & Owen, F. N. 2014, "Star Formation Rate and Extinction in Faint z ~4 Lyman Break Galaxies", *The Astrophysical Journal*, 792, 139

- **Supporting publications:**

26. Shin, T.-. hyeon., Mandelbaum, R., To, C.-H., et al. 2026, "Combining Rubin and Roman photometry and its implications for photometric redshift estimation", *arXiv e-prints*, arXiv:2610.22547
Contributed project idea, reviews, and coordination

25. Dalal, N., Hirata, C., Shiveshwarkar, C., et al. 2026, "PSFSim: PhySics First Simulations for the Point Spread Function of the Roman Space Telescope", *arXiv e-prints*, arXiv:2609.28821
Contributed code development and reviews

24. Graham, K., Jeltema, T. E., To, C.-H., et al. 2026, "Probing Projection Effects in Optically-Selected Clusters with Velocity Dispersions Outliers", *arXiv e-prints*, arXiv:2609.08095
Contributed Cardinal simulation.

23. Yin, B., Amon, A., Campos, A., et al. 2025, "Dark Energy Survey Year 6 Results: Redshift Calibration of the Weak Lensing Source Galaxies", *arXiv e-prints*, arXiv:2510.23566
Contributed Cardinal simulation-related data vector.

22. d'Assignies, W., Bernstein, G. M., Yin, B., et al. 2025, "Dark Energy Survey Year 6 Results: Clustering-redshifts and importance sampling of Self-Organised-Maps $n(z)$ realizations for 3×2pt samples", *arXiv e-prints*, arXiv:2510.23565
Contributed Cardinal simulation-related data vector.

21. DES Collaboration, Abbott, T. M. C., Adamow, M., et al. 2026, "Dark Energy Survey Year 6 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing", *arXiv e-prints*, arXiv:2601.14559
Contributed Cardinal simulation-related data vector.

20. Sanchez-Cid, D., Ferté, A., Blazek, J., et al. 2026, "Dark Energy Survey Year 6 results: Weak lensing and galaxy clustering cosmological analysis framework", *Physical Review D*, 114, 063506
Contributed Cardinal simulation-related data vector.

19. de Souza, D. H. F., Yin, B., Eifler, T., et al. 2026, "Modeling redshift uncertainties in Roman weak lensing cosmology", *Journal of Cosmology and Astroparticle Physics*,

2026, 105
Produced simulation products.

18. Costanzi, M., Wu, H. Y., Esteves, J. H., Grandis, S., To, C., & Aguena, M. 2026, "Forward analytical model for the optical selection bias on galaxy cluster lensing profiles", *arXiv e-prints*, arXiv:2604.05833
Provide expertise on projection effect modeling.

17. Yang, L., Wu, H.-Y., Jeltema, T., To, C.-H., Cawthon, R., & Cao, S. 2026, "Calibrating optical galaxy cluster projection effects with sparse spectroscopic samples: A clustering redshift approach", *arXiv e-prints*, arXiv:2604.00420
Contributed Cardinal simulation.

16. Hsu, Y.-H., Gruen, D., Gallardo, P. A., Dolag, K., To, C.-H., Wu, H.-Y., Marini, I., & Rozo, E. 2026, "Robustness of Pairwise Kinematic SZ Effect to Optical Cluster Selection Bias", *Astronomy and Astrophysics*, 712, A1
Providing expertise on simulating optical clusters in hydro-simulations

15. DES Collaboration, Cao, S., Wu, H.-Y., et al. 2025, "Association between optically identified galaxy clusters and the underlying dark matter halos", *arXiv e-prints*, arXiv:2506.17526
Contributed Cardinal simulation.

14. Pandey, S., Salcido, J., To, C.-H., Hill, J. C., Anbajagane, D., Baxter, E. J., & McCarthy, I. G. 2025, "Accurate connected modeling of gas thermodynamics and matter distribution", *Physical Review D*, 111, 043529

    Contributed to code development.

13. Salcedo, A. N., Wu, H.-Y., Rozo, E., Weinberg, D. H., To, C.-H., Sunayama, T., & Lee, A. 2024, "Consistency of Dark Energy Survey Year 1 Galaxy Clusters with Planck", *Physical Review Letters*, 133, 221002

    Contributed to project conceptualization and provided expertise on DES cluster cosmology.

12. Zhou, C., Wu, H.-Y., Salcedo, A. N., et al. 2024, "Forecasting the constraints on optical selection bias and projection effects of galaxy cluster lensing with multiwavelength data", *Physical Review D*, 110, 103508

    Contributed to project conceptualization.

11. Lee, A., Wu, H.-Y., Salcedo, A. N., et al. 2025, "Optical galaxy cluster mock catalogs with realistic projection effects: Validations with the SDSS clusters", *Physical Review D*, 111, 063502

    Contributed to project conceptualization.

10. Zaborowski, E. A., Taylor, P., Honscheid, K., et al. 2025, "A sound horizon-free measurement of H_0 in DESI 2024", *Journal of Cosmology and Astroparticle Physics*, 2025, 020

    Contributed to project conceptualization.

9. Taylor, P. L., Cuceu, A., To, C.-H., & Zaborowski, E. A. 2024, "CombineHarvester-Flow: Joint Probe Analysis Made Easy with Normalizing Flows", *The Open Journal of Astrophysics*, 7, 86

   Contributed to code development.

8. Schiappucci, E., Raghunathan, S., To, C., et al. 2025, "Constraining cosmological parameters using the pairwise kinematic Sunyaev-Zel'dovich effect with CMB-S4 and future galaxy cluster surveys", *Physical Review D*, 111, 063541

   Contributed to simulations of projection effect.

7. Zhang, Z., Wu, H.-Y., Zhang, Y., et al. 2023, "Modelling galaxy cluster triaxiality in stacked cluster weak lensing analyses", *Monthly Notices of the Royal Astronomical Society*, 523, 1994

   Contributed to project conceptualization and provided expertise on simulation products usage.

6. Zhang, T., Chuang, C.-H., Wechsler, R. H., et al. 2023, "Covariance matrices for variance-suppressed simulations", *Monthly Notices of the Royal Astronomical Society*, 518, 3737

   Contributed to project conceptualization.

5. Wu, H.-Y., Costanzi, M., To, C.-H., et al. 2022, "Optical selection bias and projection effects in stacked galaxy cluster weak lensing", *Monthly Notices of the Royal Astronomical Society*, 515, 4471

   Contributed to project conceptualization, provided expertise on projection effect, and supported simulation products usage.

4. Myles, J., Gruen, D., Mantz, A. B., et al. 2021, "Spectroscopic quantification of projection effects in the SDSS redMaPPer galaxy cluster catalogue", *Monthly Notices of the Royal Astronomical Society*, 505, 33

   Contributed to project conceptualization, provided expertise on projection effects, and derived projection effect estimators.

3. Abbott, T. M. C., Aguena, M., Alarcon, A., et al. 2020, "Dark Energy Survey Year 1 Results: Cosmological constraints from cluster abundances and weak lensing", *Physical Review D*, 102, 023509

   Helped quantify the selection effect of redMaPPer clusters and fixed bugs in the projection effect model.

2. Korytov, D., Hearin, A., Kovacs, E., et al. 2019, "CosmoDC2: A Synthetic Sky Catalog for Dark Energy Science with LSST", *The Astrophysical Journal Supplement Series*, 245, 26

   Contributed to the validation of cosmoDC2 by running redMaPPer on the catalog and by investigating the colors of red galaxies in mocks.

1. Chuang, C.-H., Yepes, G., Kitaura, F.-S., et al. 2019, "UNIT project: Universe N-body simulations for the Investigation of Theoretical models from galaxy surveys", *Monthly Notices of the Royal Astronomical Society*, 487, 48

   Derived the estimator quantifying the improvement in suppressed variance methods.

- **Selected publications that are credited as a DES builder**:

86. Kornoelje, K., Bleem, L. E., Rykoff, E. S., et al. 2025, "The SPT-Deep Cluster Catalog: Sunyaev-Zel'dovich Selected Clusters from Combined SPT-3G and SPTpol Measurements over 100 Square Degrees", *arXiv e-prints*, arXiv:2503.17271

85. DES Collaboration, Abbott, T. M. C., Acevedo, M., et al. 2025, "Dark Energy Survey: implications for cosmological expansion models from the final DES Baryon Acoustic Oscillation and Supernova data", *arXiv e-prints*, arXiv:2503.06712

84. Singh, A., Mohr, J. J., Davies, C. T., et al. 2025, "Galaxy cluster matter profiles: I. Self-similarity, mass calibration, and observable-mass relation validation employing cluster mass posteriors", *Astronomy and Astrophysics*, 695, A49

83. Anbajagane, D., Chang, C., Drlica-Wagner, A., et al. 2025, "The DECADE cosmic shear project IV: cosmological constraints from 107 million galaxies across 5,400 $\deg^2$ of the sky", *arXiv e-prints*, arXiv:2502.17677

82. Anbajagane, D., Chang, C., Chicoine, N., et al. 2025, "The DECADE cosmic shear project III: validation of analysis pipeline using spatially inhomogeneous data", *arXiv e-prints*, arXiv:2502.17676

81. Grandis, S., Costanzi, M., Mohr, J. J., et al. 2025, "Selection Function of Clusters in Dark Energy Survey Year 3 Data from Cross-Matching with South Pole Telescope Detections", *arXiv e-prints*, arXiv:2502.12914

80. McCullough, J., Amon, A., Legnani, E., et al. 2024, "Dark Energy Survey Year 3: Blue Shear", *arXiv e-prints*, arXiv:2410.22272

79. Bocquet, S., Grandis, S., Bleem, L. E., et al. 2024, "SPT clusters with DES and HST weak lensing. II. Cosmological constraints from the abundance of massive halos", *Physical Review D*, 110, 083510

78. Bocquet, S., Grandis, S., Bleem, L. E., et al. 2024, "SPT clusters with DES and HST weak lensing. I. Cluster lensing and Bayesian population modeling of multi-wavelength cluster datasets", *Physical Review D*, 110, 083509

77. Chicoine, N., Prat, J., Zacharegkas, G., et al. 2024, "Weak Gravitational Lensing around Low Surface Brightness Galaxies in the DES Year 3 Data", *The Open Journal of Astrophysics*, 7, 89

76. Klein, M., Mohr, J. J., Bocquet, S., et al. 2024, "VizieR Online Data Catalog: SPT-SZ MCMF catalog (Klein+, 2024)", *VizieR Online Data Catalog*, 753, J/MNRAS/531/3973

75. Lokken, M., van Engelen, A., Aguena, M., et al. 2024, "Superclustering with the Atacama Cosmology Telescope and Dark Energy Survey: II. Anisotropic large-scale coherence in hot gas, galaxies, and dark matter", *arXiv e-prints*, arXiv:2409.04535

74. Mena-Fernández, J., Rodríguez-Monroy, M., Avila, S., et al. 2024, "Dark Energy Survey: Galaxy sample for the baryonic acoustic oscillation measurement from the final dataset", *Physical Review D*, 110, 063514

73. Kelly, P. M., Jobel, J., Eiger, O., et al. 2024, "Dark energy survey year 3 results: miscentring calibration and X-ray-richness scaling relations in redMaPPer clusters", *Monthly Notices of the Royal Astronomical Society*, 533, 572

72. Klein, M., Mohr, J. J., Bocquet, S., et al. 2024, "SPT-SZ MCMF: an extension of the SPT-SZ catalogue over the DES region", *Monthly Notices of the Royal Astronomical Society*, 531, 3973

71. Zhang, Y., Golden-Marx, J. B., Ogando, R. L. C., et al. 2024, "Dark Energy Survey Year 6 results: Intra-cluster light from redshift 0.2 to 0.5", *Monthly Notices of the Royal Astronomical Society*, 531, 510

70. Cross, D., Thoron, G., Jeltema, T. E., et al. 2024, "Examining the self-interaction of dark matter through central cluster galaxy offsets", *Monthly Notices of the Royal Astronomical Society*, 529, 52

69. Bleem, L. E., Klein, M., Abbot, T. M. C., et al. 2024, "Galaxy Clusters Discovered via the Thermal Sunyaev-Zel'dovich Effect in the 500-square-degree SPTpol Survey", *The Open Journal of Astrophysics*, 7, 13

68. Anbajagane, D., Chang, C., Baxter, E. J., et al. 2024, "Cosmological shocks around galaxy clusters: a coherent investigation with DES, SPT, and ACT", *Monthly Notices of the Royal Astronomical Society*, 527, 9378

67. Giannini, G., Alarcon, A., Gatti, M., et al. 2024, "Dark Energy Survey Year 3 results: redshift calibration of the MAGLIM lens sample from the combination of SOMPZ and clustering and its impact on cosmology", *Monthly Notices of the Royal Astronomical Society*, 527, 2010

66. Marques, G. A., Madhavacheril, M. S., Darwish, O., et al. 2024, "Cosmological constraints from the tomography of DES-Y3 galaxies with CMB lensing from ACT DR4", *Journal of Cosmology and Astroparticle Physics*, 2024, 033

65. Zhou, C., Tong, A., Troxel, M. A., et al. 2023, "The intrinsic alignment of red galaxies in DES Y1 redMaPPer galaxy clusters", *Monthly Notices of the Royal Astronomical Society*, 526, 323

64. Dark Energy Survey and Kilo-Degree Survey Collaboration, Abbott, T. M. C., Aguena, M., et al. 2023, "DES Y3 + KiDS-1000: Consistent cosmology combining cosmic shear surveys", *The Open Journal of Astrophysics*, 6, 36

63. Samuroff, S., Mandelbaum, R., Blazek, J., et al. 2023, "The Dark Energy Survey Year 3 and eBOSS: constraining galaxy intrinsic alignments across luminosity and colour space", *Monthly Notices of the Royal Astronomical Society*, 524, 2195

62. Elvin-Poole, J., MacCrann, N., Everett, S., et al. 2023, "Dark Energy Survey Year 3 results: magnification modelling and impact on cosmological constraints from galaxy clustering and galaxy-galaxy lensing", *Monthly Notices of the Royal Astronomical Society*, 523, 3649

61. Mallaby-Kay, M., Amodeo, S., Hill, J. C., et al. 2023, "Kinematic Sunyaev-Zel'dovich effect with ACT, DES, and BOSS: A novel hybrid estimator", *Physical Review D*, 108, 023516

60. Upsdell, E. W., Giles, P. A., Romer, A. K., et al. 2023, "The XMM cluster survey: exploring scaling relations and completeness of the dark energy survey year 3 redMaPPer cluster catalogue", *Monthly Notices of the Royal Astronomical Society*, 522, 5267

59. Prat, J., Zacharegkas, G., Park, Y., et al. 2023, "Non-local contribution from small scales in galaxy-galaxy lensing: comparison of mitigation schemes", *Monthly Notices of the Royal Astronomical Society*, 522, 412

58. Lemos, P., Weaverdyck, N., Rollins, R. P., et al. 2023, "Robust sampling for weak lensing and clustering analyses with the Dark Energy Survey", *Monthly Notices of the Royal Astronomical Society*, 521, 1184

57. Abbott, T. M. C., Aguena, M., Alarcon, A., et al. 2023, "Dark Energy Survey Year 3 results: Constraints on extensions to Λ CDM with weak lensing and galaxy clustering", *Physical Review D*, 107, 083504

56. Schiappucci, E., Bianchini, F., Aguena, M., et al. 2023, "Measurement of the mean central optical depth of galaxy clusters via the pairwise kinematic Sunyaev-Zel'dovich effect with SPT-3G and DES", *Physical Review D*, 107, 042004

55. Chen, A., Aricò, G., Huterer, D., et al. 2023, "Constraining the baryonic feedback with cosmic shear using the DES Year-3 small-scale measurements", *Monthly Notices of the Royal Astronomical Society*, 518, 5340

54. Abbott, T. M. C., Aguena, M., Alarcon, A., et al. 2023, "Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck. III. Combined cosmological constraints", *Physical Review D*, 107, 023531

53. Chang, C., Omori, Y., Baxter, E. J., et al. 2023, "Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and P l a n c k . II. Cross-correlation measurements and cosmological constraints", *Physical Review D*, 107, 023530

52. Omori, Y., Baxter, E. J., Chang, C., et al. 2023, "Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck. I. Construction of CMB lensing maps and modeling choices", *Physical Review D*, 107, 023529

51. Amon, A., Robertson, N. C., Miyatake, H., et al. 2023, "Consistent lensing and clustering in a low-S_8 Universe with BOSS, DES Year 3, HSC Year 1, and KiDS-1000", *Monthly Notices of the Royal Astronomical Society*, 518, 477

50. Porredon, A., Crocce, M., Elvin-Poole, J., et al. 2022, "Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and galaxy-galaxy lensing using the MAGLIM lens sample", *Physical Review D*, 106, 103530

49. Gatti, M., Jain, B., Chang, C., et al. 2022, "Dark Energy Survey Year 3 results: Cosmology with moments of weak lensing mass maps", *Physical Review D*, 106, 083509

48. Chen, R., Scolnic, D., Rozo, E., et al. 2022, "Measuring Cosmological Parameters with Type Ia Supernovae in redMaGiC Galaxies", *The Astrophysical Journal*, 938, 62

47. Kovács, A., Vielzeuf, P., Ferrero, I., et al. 2022, "Dark Energy Survey Year 3 results: Imprints of cosmic voids and superclusters in the Planck CMB lensing map", *Monthly Notices of the Royal Astronomical Society*, 515, 4417

46. Pandey, S., Krause, E., DeRose, J., et al. 2022, "Dark Energy Survey year 3 results: Constraints on cosmological parameters and galaxy-bias models from galaxy clustering and galaxy-galaxy lensing using the redMaGiC sample", *Physical Review D*, 106, 043520

45. Lokken, M., Hložek, R., van Engelen, A., et al. 2022, "Superclustering with the Atacama Cosmology Telescope and Dark Energy Survey. I. Evidence for Thermal Energy Anisotropy Using Oriented Stacking", *The Astrophysical Journal*, 933, 134

44. Pandey, S., Gatti, M., Baxter, E., et al. 2022, "Cross-correlation of Dark Energy Survey Year 3 lensing data with ACT and P l a n c k thermal Sunyaev-Zel'dovich effect observations. II. Modeling and constraints on halo pressure profiles", *Physical Review D*, 105, 123526

43. Gatti, M., Pandey, S., Baxter, E., et al. 2022, "Cross-correlation of Dark Energy Survey Year 3 lensing data with ACT and Planck thermal Sunyaev-Zel'dovich effect observations. I. Measurements, systematics tests, and feedback model constraints", *Physical Review D*, 105, 123525

42. DeRose, J., Wechsler, R. H., Becker, M. R., et al. 2022, "Dark Energy Survey Year 3 results: Cosmology from combined galaxy clustering and lensing validation on cosmological simulations", *Physical Review D*, 105, 123520

41. Secco, L. F., Jarvis, M., Jain, B., et al. 2022, "Dark Energy Survey Year 3 Results: Three-point shear correlations and mass aperture moments", *Physical Review D*, 105, 103537

40. Sánchez, C., Prat, J., Zacharegkas, G., et al. 2022, "Dark Energy Survey Year 3 results: Exploiting small-scale information with lensing shear ratios", *Physical Review D*, 105, 083529

39. Prat, J., Blazek, J., Sánchez, C., et al. 2022, "Dark energy survey year 3 results: High-precision measurement and modeling of galaxy-galaxy lensing", *Physical Review D*, 105, 083528

38. Rodríguez-Monroy, M., Weaverdyck, N., Elvin-Poole, J., et al. 2022, "Dark Energy Survey Year 3 results: galaxy clustering and systematics treatment for lens galaxy samples", *Monthly Notices of the Royal Astronomical Society*, 511, 2665

37. Cordero, J. P., Harrison, I., Rollins, R. P., et al. 2022, "Dark Energy Survey Year 3 results: marginalization over redshift distribution uncertainties using ranking of discrete realizations", *Monthly Notices of the Royal Astronomical Society*, 511, 2170

36. Somboonpanyakul, T., McDonald, M., Noble, A., et al. 2022, "The Evolution of AGN Activity in Brightest Cluster Galaxies", *The Astronomical Journal*, 163, 146

35. Leauthaud, A., Amon, A., Singh, S., et al. 2022, "Lensing without borders - I. A blind comparison of the amplitude of galaxy-galaxy lensing between independent imaging surveys", *Monthly Notices of the Royal Astronomical Society*, 510, 6150

34. Abbott, T. M. C., Aguena, M., Allam, S., et al. 2022, "Dark Energy Survey Year 3 results: A 2.7% measurement of baryon acoustic oscillation distance scale at redshift 0.835", *Physical Review D*, 105, 043512

33. Gatti, M., Giannini, G., Bernstein, G. M., et al. 2022, "Dark Energy Survey Year 3 Results: clustering redshifts - calibration of the weak lensing source redshift distributions with redMaGiC and BOSS/eBOSS", *Monthly Notices of the Royal Astronomical Society*, 510, 1223

32. Lee, S., Huff, E. M., Choi, A., et al. 2022, "Probing gravity with the DES-CMASS sample and BOSS spectroscopy", *Monthly Notices of the Royal Astronomical Society*, 509, 4982

31. Varga, T. N., Gruen, D., Seitz, S., et al. 2022, "Synthetic galaxy clusters and observations based on Dark Energy Survey Year 3 Data", *Monthly Notices of the Royal Astronomical Society*, 509, 4865

30. Abbott, T. M. C., Aguena, M., Alarcon, A., et al. 2022, "Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing", *Physical Review D*, 105, 023520

29. Secco, L. F., Samuroff, S., Krause, E., et al. 2022, "Dark Energy Survey Year 3 results: Cosmology from cosmic shear and robustness to modeling uncertainty", *Physical Review D*, 105, 023515

28. Amon, A., Gruen, D., Troxel, M. A., et al. 2022, "Dark Energy Survey Year 3 results: Cosmology from cosmic shear and robustness to data calibration", *Physical Review D*, 105, 023514

27. MacCrann, N., Becker, M. R., McCullough, J., et al. 2022, "Dark Energy Survey Y3 results: blending shear and redshift biases in image simulations", *Monthly Notices of the Royal Astronomical Society*, 509, 3371

26. Zacharegkas, G., Chang, C., Prat, J., et al. 2022, "Dark Energy Survey Year 3 results: galaxy-halo connection from galaxy-galaxy lensing", *Monthly Notices of the Royal Astronomical Society*, 509, 3119

25. Lee, S., Troxel, M. A., Choi, A., et al. 2022, "Galaxy-galaxy lensing with the DES-CMASS catalogue: measurement and constraints on the galaxy-matter cross-correlation", *Monthly Notices of the Royal Astronomical Society*, 509, 2033

24. Carnero Rosell, A., Rodriguez-Monroy, M., Crocce, M., et al. 2022, "Dark Energy Survey Year 3 results: galaxy sample for BAO measurement", *Monthly Notices of the Royal Astronomical Society*, 509, 778

23. Everett, S., Yanny, B., Kuropatkin, N., et al. 2022, "Dark Energy Survey Year 3 Results: Measuring the Survey Transfer Function with Balrog", *The Astrophysical Journal Supplement Series*, 258, 15

22. Friedrich, O., Andrade-Oliveira, F., Camacho, H., et al. 2021, "Dark Energy Survey year 3 results: covariance modelling and its impact on parameter estimation and quality of fit", *Monthly Notices of the Royal Astronomical Society*, 508, 3125

21. Adhikari, S., Shin, T.-. hyeon., Jain, B., et al. 2021, "Probing Galaxy Evolution in Massive Clusters Using ACT and DES: Splashback as a Cosmic Clock", *The Astrophysical Journal*, 923, 37

20. Shin, T., Jain, B., Adhikari, S., et al. 2021, "The mass and galaxy distribution around SZ-selected clusters", *Monthly Notices of the Royal Astronomical Society*, 507, 5758

19. Fortino, W. F., Bernstein, G. M., Bernardinelli, P. H., et al. 2021, "Reducing Ground-based Astrometric Errors with Gaia and Gaussian Processes", *The Astronomical Journal*, 162, 106

18. Lemos, P., Raveri, M., Campos, A., et al. 2021, "Assessing tension metrics with dark energy survey and Planck data", *Monthly Notices of the Royal Astronomical Society*, 505, 6179

17. Jeffrey, N., Gatti, M., Chang, C., et al. 2021, "Dark Energy Survey Year 3 results: Curved-sky weak lensing mass map reconstruction", *Monthly Notices of the Royal Astronomical Society*, 505, 4626

16. Myles, J., Alarcon, A., Amon, A., et al. 2021, "Dark Energy Survey Year 3 results: redshift calibration of the weak lensing source galaxies", *Monthly Notices of the Royal Astronomical Society*, 505, 4249

15. Grayling, M., Gutiérrez, C. P., Sullivan, M., et al. 2021, "Understanding the extreme luminosity of DES14X2fna", *Monthly Notices of the Royal Astronomical Society*, 505, 3950

14. Vincenzi, M., Sullivan, M., Graur, O., et al. 2021, "The Dark Energy Survey supernova programme: modelling selection efficiency and observed core-collapse supernova contamination", *Monthly Notices of the Royal Astronomical Society*, 505, 2819

13. Gatti, M., Sheldon, E., Amon, A., et al. 2021, "Dark energy survey year 3 results: weak lensing shape catalogue", *Monthly Notices of the Royal Astronomical Society*, 504, 4312

12. Chen, A., Huterer, D., Lee, S., et al. 2021, "Constraints on dark matter to dark radiation conversion in the late universe with DES-Y1 and external data", *Physical Review D*, 103, 123528

11. Grandis, S., Mohr, J. J., Costanzi, M., et al. 2021, "Exploring the contamination of the DES-Y1 cluster sample with SPT-SZ selected clusters", *Monthly Notices of the Royal Astronomical Society*, 504, 1253

10. Sevilla-Noarbe, I., Bechtol, K., Carrasco Kind, M., et al. 2021, "Dark Energy Survey Year 3 Results: Photometric Data Set for Cosmology", *The Astrophysical Journal Supplement Series*, 254, 24

9. Krause, E., Fang, X., Pandey, S., et al. 2021, "Dark Energy Survey Year 3 Results: Multi-Probe Modeling Strategy and Validation", *arXiv e-prints*, arXiv:2105.13548

8. Doux, C., Chang, C., Jain, B., et al. 2021, "Consistency of cosmic shear analyses in harmonic and real space", *Monthly Notices of the Royal Astronomical Society*, 503, 3796

7. Doux, C., Baxter, E., Lemos, P., et al. 2021, "Dark energy survey internal consistency tests of the joint cosmological probes analysis with posterior predictive distributions", *Monthly Notices of the Royal Astronomical Society*, 503, 2688

6. Aguena, M., Benoist, C., da Costa, L. N., et al. 2021, "The WaZP galaxy cluster sample of the dark energy survey year 1", *Monthly Notices of the Royal Astronomical Society*, 502, 4435

5. Hilton, M., Sifón, C., Naess, S., et al. 2021, "The Atacama Cosmology Telescope: A Catalog of 4000 Sunyaev-Zel'dovich Galaxy Clusters", *The Astrophysical Journal Supplement Series*, 253, 3

4. Costanzi, M., Saro, A., Bocquet, S., et al. 2021, "Cosmological constraints from DES Y1 cluster abundances and SPT multiwavelength data", *Physical Review D*, 103, 043522

3. Porredon, A., Crocce, M., Fosalba, P., et al. 2021, "Dark Energy Survey Year 3 results: Optimizing the lens sample in a combined galaxy clustering and galaxy-galaxy lensing analysis", *Physical Review D*, 103, 043503

2. Muir, J., Baxter, E., Miranda, V., et al. 2021, "DES Y1 results: Splitting growth and geometry to test Λ CDM", *Physical Review D*, 103, 023528

1. Zhang, Y., Yanny, B., Palmese, A., et al. 2019, "Dark Energy Survey Year 1 Results: Detection of Intracluster Light at Redshift ~ 0.25", *The Astrophysical Journal*, 874, 165

REFERENCES

- Chihway Chang Email: chihway@kicp.uchicago.edu
  William Eckhardt Research Center
  5640 S Ellis Ave, Chicago, IL, 60637

- Elisabeth Krause Email: krausee@arizona.edu
  Steward Observatory
  Department of Astronomy, University of Arizona
  Tucson, AZ 85721, USA

- Risa H. Wechsler Email: rwechsler@stanford.edu
  Physics Department, Stanford University
  Stanford, CA 94305, USA

- David Weinberg Email: weinberg.21@osu.edu
  4055 McPherson Laboratory
  140 West 18th Avenue Columbus, Ohio 43210-1173